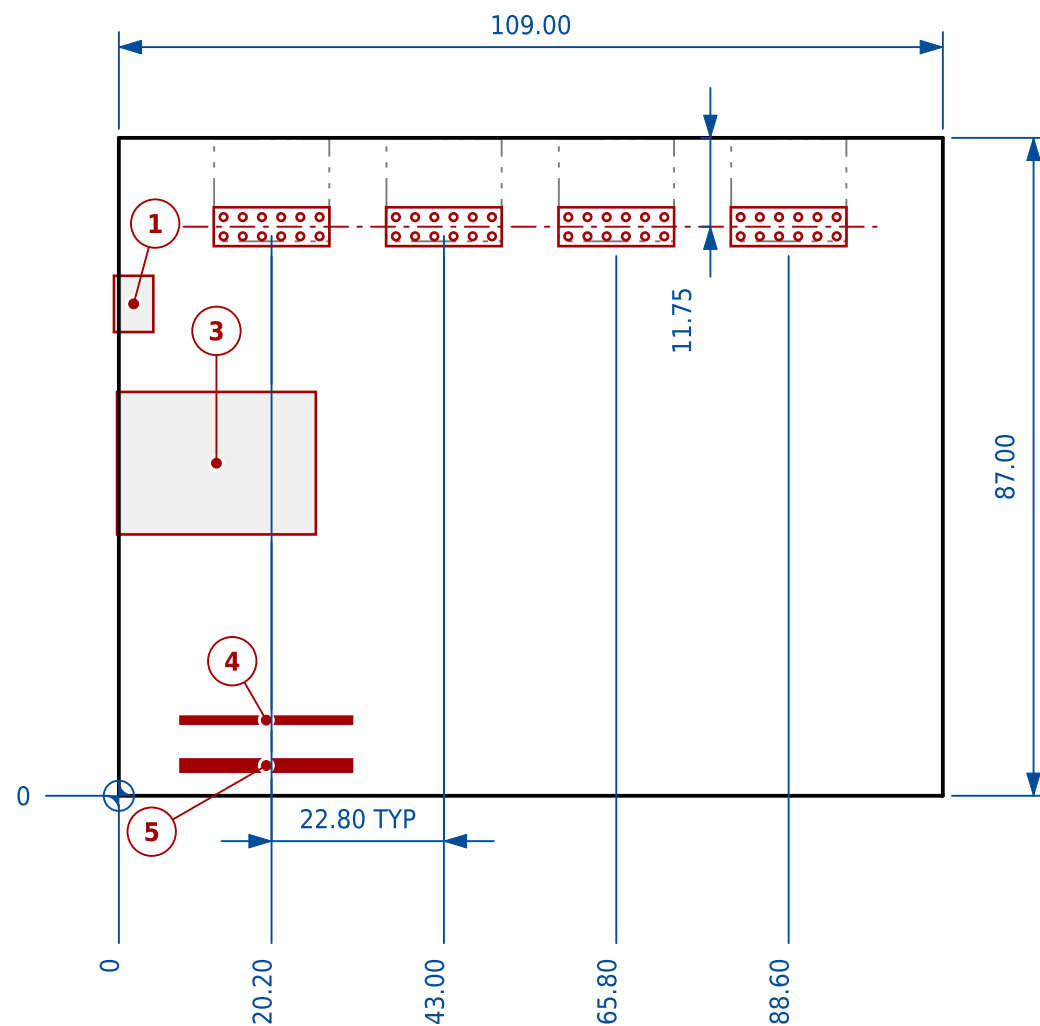




NOTES

1. Viewed from the component side.
2. No mounting holes. The drawing shows four 10.0 mm rubber feet centred 5.0 mm from each corner; a plate has to carry the board on those or clamp its edges.
3. The Pmod pin rows are 2.50 mm apart on the drawing, not 2.54, and the hosts are on a 22.80 mm pitch where the PDF's dimension reads 0.90 in; both within the general tolerance.
4. JA is nearest the USB and Ethernet corner, per the reference manual's callout figure.
5. The 1.6 x 1.6 mm LED row nearest the edge is taken as the tri-colour LD0-LD3 and the 0603 row behind it as LD4-LD7; no source names them.
6. Bodies read from the PDF plot are good to about +/-0.3 mm; the DXF figures carry the general tolerance.
7. Assembled envelope, connector overhang included: 109.65 x 87.00 mm.

SOURCES

1. Mechanical drawing:
https://digilent.com/reference/_media/reference/programmable-logic/arti-a7/arti_a7.zip - Digilent, Arty_A7_DXF.DXF and Mechanical_Arty_A7.pdf, dated 2020-09-02. Outline, pin fields, slots and pegs from the DXF; component bodies from the PDF, whose plot scale is recovered from the outline (axes agree to 0.20 %).
2. 3D model: https://digilent.com/reference/_media/reference/programmable-logic/arti/arti_rev_cad.zip - Digilent, Arty Rev C: used only to identify which of the PDF's outlines are the eight user LEDs.
3. Reference manual:
<https://digilent.com/reference/programmable-logic/arti-a7/reference-manual> - Pmod JA, JB, JC, JD; four tri-colour and four single LEDs.
4. Mounting:
<https://forum.digilent.com/topic/4523-410-319-mounting/> - Digilent staff, 2017-07-19: "There are no thru holes or mounting provisions"; the board stands on rubber feet.

PMOD HOST HEADERS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	top	20.200	75.250	13.850	74.000
JB	top	43.000	75.250	36.650	74.000
JC	top	65.800	75.250	59.450	74.000
JD	top	88.600	75.250	82.250	74.000

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB J10, programming and console	-0.65 to 4.56	61.32 to 68.76
2	Second USB port - not on this board	-	-
3	Ethernet RJ45 J9	-0.24 to 26.06	34.56 to 53.40
4	LEDs LD4-LD7, 4 on 7.00 mm pitch	8.16 to 30.84	9.54 to 10.46
5	Tri-colour LEDs LD0-LD3, 4 on 7.00 mm pitch	8.16 to 30.84	3.19 to 4.80
6	Expansion connector, first - not on this board	-	-
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-

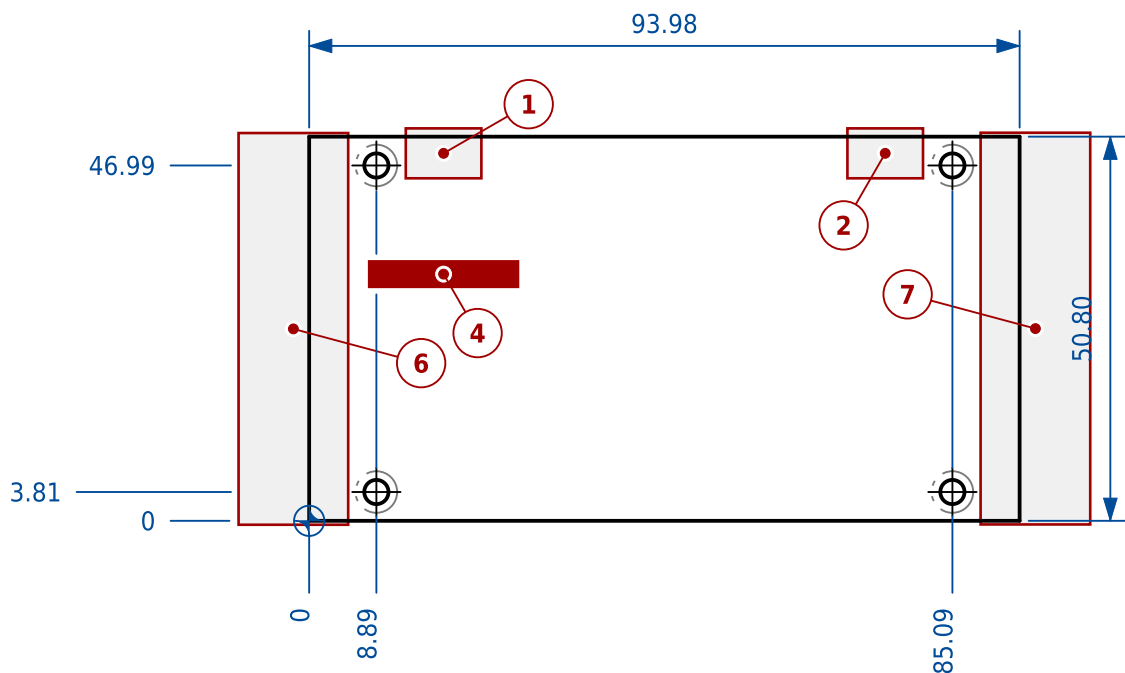
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

DIGILENT ARTY A7

A7-35T and A7-100T

DRAWN generated		VERSION v0.0-186-gd944fb5	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-ARTY-A7			REV A
MATERIAL PCB, thickness not stated		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	8.89	3.81	3.20	5.50
H2	85.09	3.81	3.20	5.50
H3	85.09	46.99	3.20	5.50
H4	8.89	46.99	3.20	5.50

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB US1, FTDI programming and console	12.78 to 22.78	45.30 to 51.90
2	micro-USB US2, USB host / OTG	71.20 to 81.20	45.30 to 51.90
3	Ethernet RJ45 - not on this board	-	-
4	LEDs D0-D7, 8 on 2.54 mm pitch	7.94 to 27.62	30.94 to 34.30
5	User LEDs, tri-colour - not on this board	-	-
6	GPIO header J1, 2x20	-9.34 to 5.16	-0.52 to 51.28
7	GPIO header J2, 2x20	88.82 to 103.32	-0.48 to 51.32
8	Expansion connector, third - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- No Pmod host. J1 and J2 are right-angle 2x20 sockets on 2.54 mm whose pinout is Pmod compatible, per the maker; pin 1 of each is given in the schedule note.
- No Ethernet jack.
- Assembled envelope, connector overhang included: 112.66 x 52.42 mm.

SOURCES

- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ 300a361b (tag v3.0.3) - Radiona / emard
- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ d518b2ce (tag v3.0.7) - Radiona / emard

SOURCES (continued)

- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ 9908a1f8 (tag v3.0.8) - Radiona / emard
- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ cc1817d2 (tag v3.1.7) - Radiona / emard
- Board versions: <https://github.com/emard/ulx3s/blob/master/doc/MANUAL.md> - section "Board Versions": v3.0.3, v3.0.7, v3.0.8 and v3.1.7 are the rows marked "for sale"; the four board files are read and agree on everything drawn here.
- Board size: <https://github.com/emard/ulx3s/blob/master/README.md> - quoted: "This is a small (94x51 mm) standalone FPGA board"; the board file gives 93.98 x 50.80, which is 37 x 20 tenths of an inch.

ULX3S

v3.0.3, v3.0.7, v3.0.8 and v3.1.7

DRAWN generated		VERSION v0.0-186-gd944fb5	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-ULX3S			REV A
MATERIAL PCB, 1.6 nominal		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			

A

B

C

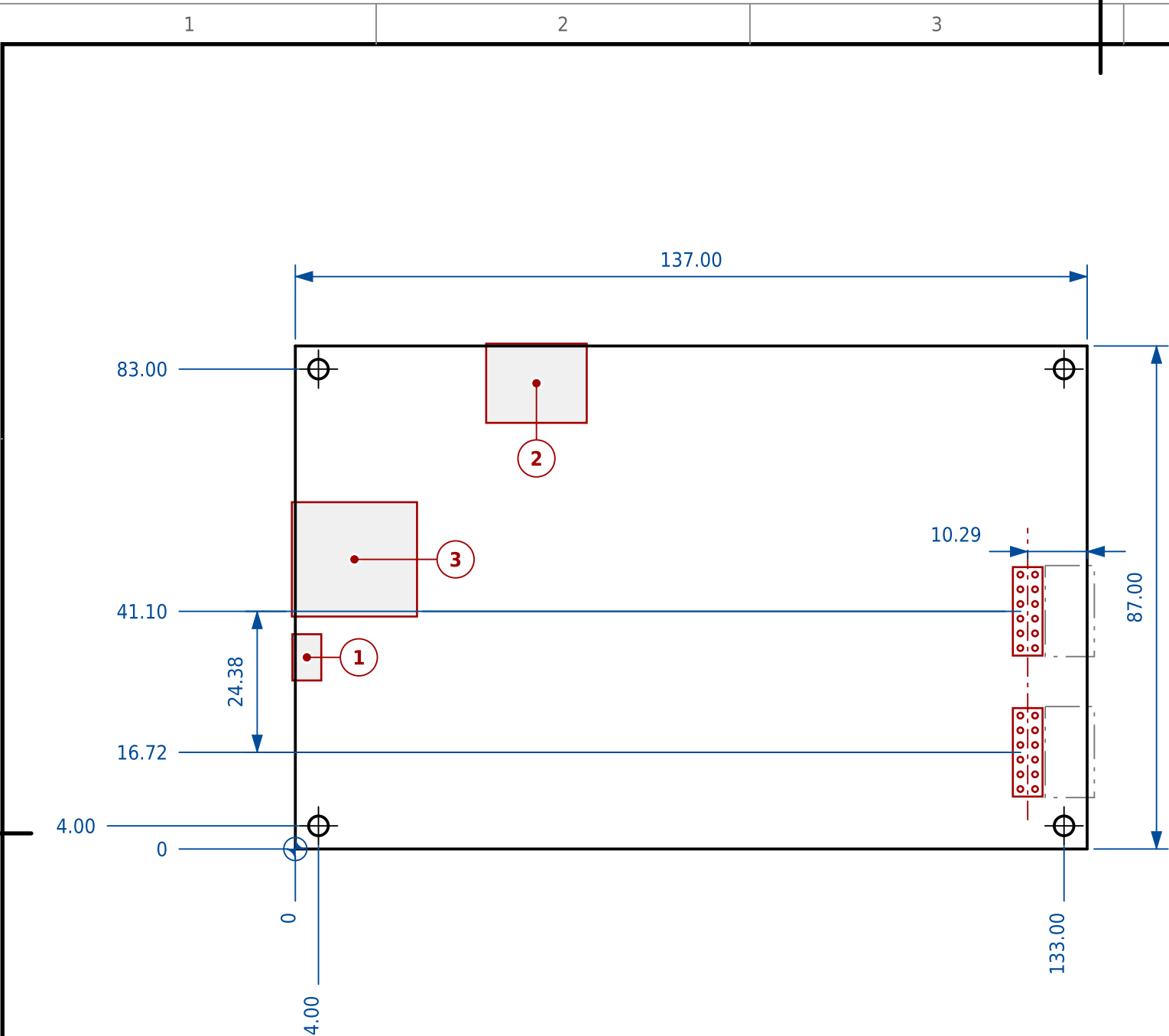
D

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	4.00	3.40	not given
MT2	133.00	4.00	3.40	not given
MT3	4.00	83.00	3.40	not given
MT4	133.00	83.00	3.40	not given

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
Pmod A	JA	right	126.713	41.101	125.443	47.451
Pmod B	JB	right	126.713	16.717	125.443	23.067

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB, programming and console	-0.51 to 4.49	29.15 to 37.15
2	USB type A host port	33.02 to 50.42	73.70 to 87.40
3	Ethernet RJ45	-0.60 to 21.05	40.20 to 59.98
4	User LEDs, single colour - not on this board	-	-
5	User LEDs, tri-colour - not on this board	-	-
6	Expansion connector, first - not on this board	-	-
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Every dimension is read from the STEP model; TUL publish no drawing. Hole IDs are assigned by this drawing.
- The user LEDs LD0-LD3 and the two tri-colour LEDs are not in the model, so schedule rows 4 and 5 are empty; the board has them, above the four push buttons near the lower edge.
- TUL's product sheet gives the board as 87 x 140 mm; the model's laminate is 137.0 wide and the Pmod hosts reach 1.2 beyond it.
- Assembled envelope, connector overhang included: 137.60 x 87.40 mm.

SOURCES

- 3D model:
<https://discuss.pynq.io/t/pynq-z2-pcb-3d-step-files/2645> - PYNQ_Z2_20220218.STEP, TUL's SolidWorks export of 2022-02-18 posted by Xilinx; the earlier model on the same thread has the Pmod spacing wrong and is not used.
- User manual:
https://dpoauwqwqsy2x.cloudfront.net/Download/PYNQ_Z2_User_Manual_v1.1.pdf - v1.1: Pmod A is the upper host (section 16); 4 LEDs and 2 tri-colour LEDs (section 14).

TUL PYNQ-Z2

137 x 87 mm

DRAWN generated		VERSION v0.0-186-gd944fb5		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO FPGA-PYNQ-Z2				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				

A

B

C

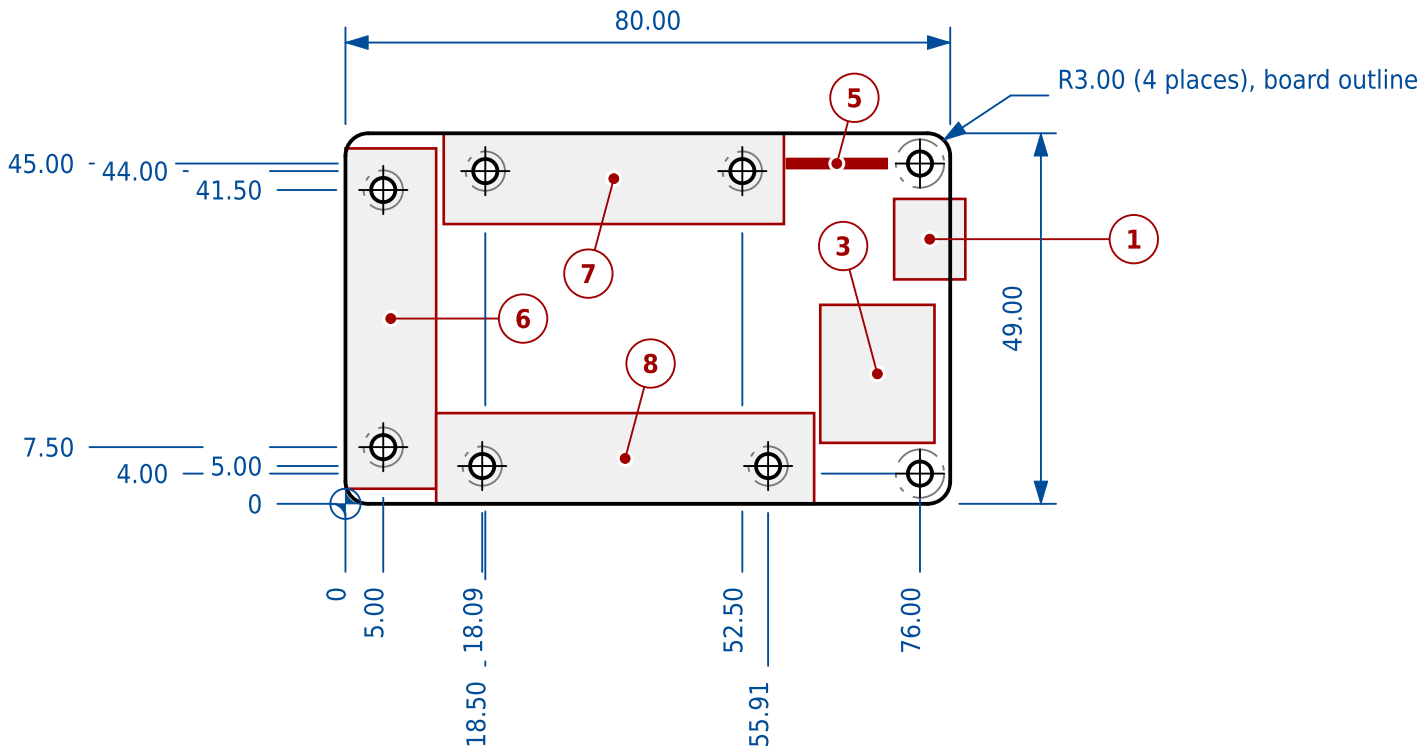
D

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	76.00	45.00	3.20	6.40
H2	76.00	4.00	3.20	6.40
SA1	5.00	7.50	3.20	5.20
SA2	5.00	41.50	3.20	5.20
SB1	18.50	44.00	3.20	5.20
SB2	52.50	44.00	3.20	5.20
SC1	18.09	5.00	3.20	5.20
SC2	55.91	5.00	3.20	5.20

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J1, programming and console	72.58 to 82.00	29.68 to 40.32
2	Second USB port - not on this board	-	-
3	Ethernet RJ45 CON1	62.81 to 77.92	8.06 to 26.32
4	User LEDs, single colour - not on this board	-	-
5	RGB LEDs D1-D7, 7 on 2.00 mm pitch	58.40 to 71.60	44.40 to 45.60
6	SYZYGY port A, J3	0.00 to 12.00	2.00 to 47.00
7	SYZYGY port B, J4	13.00 to 58.00	37.00 to 49.00
8	SYZYGY port C, J5	12.00 to 62.00	0.00 to 12.00

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs H1 and H2 are the KiCad reference designators; SA1 to SC2 are the SYZYGY standoff holes of ports A, B and C, plated, and the maker's acrylic plate bolts through all eight.
- No Pmod host: expansion is three SYZYGY ports.
- Assembled envelope, connector overhang included: 82.00 x 49.00 mm.

SOURCES

- KiCad board file:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/ButterStick_r1.0/ButterStick.kicad_pcb @ fac6a834 - title block: ButterStick rev r1.0a, dated 2021-01-10; Greg Davill / GsD
- Board size:
<https://github.com/butterstick-fpga/butterstick-hardware/blob/main/README.md> - quoted: "Board dimensions: 80mm x 49mm"
- Thickness:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/hardware/ButterStick_r1.0/Production/ButterStick-r1.0a-fab-notes.txt - quoted: "Finished board thickness is 1.6 mm."

BUTTERSTICK

r1.0			
DRAWN		VERSION	SCALE
generated		v0.0-186-gd944fb5	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	FPGA-BUTTERSTICK	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	